

2024 AMC 10A Problem 1 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 1. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 1

**Problem:**

What is the value of $101 \cdot 9,901 - 99 \cdot 10,101$?

Answer Choices:

- A. 2
- B. 20
- C. 21
- D. 200
- E. 2020

Solution:

Write the difference as

$$(100 + 1) \cdot (9900 + 1) - 99 \cdot (10,000 + 100 + 1).$$

Applying the distributive property gives

$$(990,000 + 9,900 + 100 + 1) - (990,000 + 9,900 + 99) = 100 + 1 - 99 = (A) \boxed{2}.$$

OR

Let $x = 100$. Then the minuend (the first quantity in the subtraction operation) is

$$(x + 1)(x^2 - x + 1) = x^3 + 1,$$

and the subtrahend (the quantity being subtracted from the minuend) is

$$(x - 1)(x^2 + x + 1) = x^3 - 1.$$

The difference is

$$(x^3 + 1) - (x^3 - 1) = (A)\boxed{2}.$$

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